A SHARP HARDY INEQUALITY ON $\mathbb{Z}^2$

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Abstract. We prove the Hardy inequality on $\mathbb{Z}^2$ with Euclidean weight $|x|^{-2}(\log|x|)^{-2}$ and sharp constant $1/4$, for finitely supported functions vanishing on $\{|x|\le 1\}$.

1. Introduction

We determine the sharp coefficient of the Euclidean logarithmic Hardy weight for the nearest-neighbor Laplacian on the square lattice. Throughout, $|x|$ denotes the Euclidean norm and log the natural logarithm. Let $e_1=(1,0)$, $e_2=(0,1)$, and set

$$K=\{x\in\mathbb{Z}^2:|x|\le 1\}=\{0,\pm e_1,\pm e_2\},\qquad D=\mathbb{Z}^2\setminus K.$$

We identify $C_c(D)$ with the real-valued finitely supported functions on $\mathbb{Z}^2$ that vanish on K. Their Dirichlet energy is

$$\mathcal{E}(u)=\sum_{x\in\mathbb{Z}^2}\sum_{j=1}^{2}|u(x+e_j)-u(x)|^2, \tag{1.1}$$

so that each unoriented edge is counted once.

Theorem 1.1. *Every $u\in C_c(D)$ satisfies*

$$\sum_{x\in\mathbb{Z}^2}\sum_{j=1}^{2}|u(x+e_j)-u(x)|^2\ge\frac{1}{4}\sum_{x\in D}\frac{|u(x)|^2}{|x|^2(\log|x|)^2}. \tag{1.2}$$

The constant $1/4$ is sharp, and the inequality is strict for every nonzero $u\in C_c(D)$.

Kapitanski and Laptev [3, Theorem 4.1] established a logarithmic Hardy inequality on $\mathbb{Z}^2$ with an unspecified positive coefficient. Their theorem uses the same Dirichlet set K, with the Euclidean weighted sum restricted to $|x|\ge 2$. Inequality (1.2) also includes the four points $(\pm1,\pm1)$.

Huang and Ye [2, Remark 6.2] obtained a logarithmic Hardy weight on $\mathbb{Z}^2$ with leading coefficient $1/4$, for functions supported on $\{|x|\ge 2\}$. Its next term is negative along diagonal directions, so their expansion does not establish (1.2) with the prescribed weight.

Hake, Keller and Pogorzelski [1, Corollary 2 and the subsequent remark] constructed an optimal Hardy weight from the lattice potential kernel. Its leading term is $1/[4|x|^2(\log|x|)^2]$, followed by a negative correction of order $|x|^{-2}(\log|x|)^{-3}$. They proved sharpness of the leading coefficient near infinity and discussed whether $1/4$ remains valid globally for the prescribed logarithmic weight. Theorem 1.1 establishes this endpoint for K. Their criticality result concerns the different class of functions vanishing only at the origin.

The proof uses the ground-state identity [4]. We start from the radial function $\sqrt{\frac{1}{2}\log(|x|^2-\frac{1}{2})}$. An angular comparison bounds its four-neighbor sum by the corresponding sum in a diagonal direction, where an exact factorization gives the required estimate. Changing the values on the two orbits $|x|^2=2,4$ then produces a global supersolution. The construction yields a positive remainder, while logarithmic cutoffs prove sharpness.

A direct consequence is the following family of inequalities.

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Corollary 1.2. *For each $b \geq 1$, every $u \in C_c(D)$ satisfies the sharp inequality*

$$\mathcal{E}(u) \geq \frac{1}{4} \sum_{x \in D} \frac{|u(x)|^2}{|x|^2 \log^2(b|x|)}. \tag{1.3}$$

The inequality follows from (1.2) because $\log(b|x|) \geq \log|x|$ on D; sharpness is proved in Section 3. The inequalities and their sharp constants also hold for complex-valued functions, by applying the inequalities to the real and imaginary parts.

2. The supersolution

Write $x \sim y$ when $|x - y| = 1$, and define

$$L\phi(x) = 4\phi(x) - \sum_{y \sim x} \phi(y), \qquad W(x) = \frac{1}{4|x|^2(\log|x|)^2} \quad (x \in D).$$

Thus $\mathcal{E}(u) = \sum_x u(x) Lu(x)$. Our goal is to construct $\phi > 0$ on D, with $\phi = 0$ on K, such that $L\phi > W\phi$ on D. The following identity explains this reduction.

Lemma 2.1. *If $\phi > 0$ on D and $\phi = 0$ on K, then for every $u \in C_c(D)$,*

$$\mathcal{E}(u) = \sum_{\substack{\{x,y\}: x \sim y \\ x,y \in D}} \phi(x)\phi(y) \left(\frac{u(x)}{\phi(x)} - \frac{u(y)}{\phi(y)} \right)^2 + \sum_{x \in D} \frac{L\phi(x)}{\phi(x)} u(x)^2. \tag{2.1}$$

Each edge in the first sum is counted once.

Proof. For $x, y \in D$, expansion gives

$$\begin{aligned}(u(x) - u(y))^2 = \phi(x)\phi(y) \left(\frac{u(x)}{\phi(x)} - \frac{u(y)}{\phi(y)} \right)^2 \\ + \frac{\phi(x) - \phi(y)}{\phi(x)} u(x)^2 + \frac{\phi(y) - \phi(x)}{\phi(y)} u(y)^2.\end{aligned}$$

Sum over the edges in D. An edge from $x \in D$ to K contributes $u(x)^2$ to the energy and the same amount to the potential term, since $u = \phi = 0$ on K. This proves (2.1). □

Lemma 2.2. *Let $q > 0$, and suppose that $F \in C^4$ on an open interval containing $[q + 1 - 2\sqrt{q}, q + 1 + 2\sqrt{q}]$, with $F^{(4)} \leq 0$ there. If $x_1^2 + x_2^2 = q$, then*

$$\begin{aligned}\sum_{j=1}^{2} \big(F(q + 1 + 2x_j) + F(q + 1 - 2x_j)\big) \\ \leq 2F(q + 1 + \sqrt{2q}) + 2F(q + 1 - \sqrt{2q}).\end{aligned} \tag{2.2}$$

Proof. Put $a = q + 1$ and $G(s) = F(a + 2\sqrt{s}) + F(a - 2\sqrt{s})$ for $0 \leq s \leq q$. For $s > 0$, with $t = 2\sqrt{s}$, differentiation gives

$$G''(s) = \frac{4}{t^3} \left[t\big(F''(a + t) + F''(a - t)\big) - \int_{a-t}^{a+t} F''(v)\, dv \right] \leq 0.$$

Indeed, F'' is concave and therefore lies above the chord joining its endpoint values. By continuity, G is concave on $[0, q]$. Jensen's inequality now gives $G(x_1^2) + G(x_2^2) \leq 2G(q/2)$, which is (2.2). □

We use the function

$$F(v) = \sqrt{\frac{1}{2} \log\left(v - \frac{1}{2}\right)} \qquad (v > 3/2)$$

and define

$$\phi(x) = \begin{cases} 0, & |x|^2 \le 1, \\ \dfrac{3}{5}, & |x|^2 = 2, \\ \dfrac{18}{25}, & |x|^2 = 4, \\ F(|x|^2), & |x|^2 \ge 5. \end{cases} \tag{2.3}$$

These cases exhaust $\mathbb{Z}^2$, and $\phi > 0$ on D.

Proposition 2.3. *The function (2.3) satisfies $(L-W)\phi > 0$ on D. More precisely, for $q = |x|^2 \ge 8$,*

$$\frac{L\phi(x)}{\phi(x)} \ge \frac{1}{q \log^2(q - \frac{1}{2})} > W(x), \tag{2.4}$$

and for $1 < |x|^2 < 8$, we have $(L - W)\phi(x) > 1/1250$.

Proof. Suppose first that $q \ge 8$. Then $\phi(x) = F(q)$ and $\phi(y) \le F(|y|^2)$ for every neighbor y of x. Indeed, a neighbor can meet a modified orbit only when $q = 9$ and $|y|^2 = 4$, and $18/25 < F(4)$, as verified below. Writing $t = \log(v - \frac{1}{2}) > 0$, we have

$$F^{(4)}(v) = -\frac{48t^3 + 44t^2 + 36t + 15}{16\sqrt{2}\,(v - \frac{1}{2})^4 t^{7/2}} < 0.$$

Since $(\sqrt{q} - 1)^2 \ge (\sqrt{8} - 1)^2 > 3/2$, Lemma 2.2 applies to the four neighboring values. Set

$$h = \log\left(q - \frac{1}{2}\right), \qquad d = \log\frac{\sqrt{2q} + 1}{\sqrt{2q} - 1}.$$

The factorization $q + \frac{1}{2} \pm \sqrt{2q} = \frac{1}{2}(\sqrt{2q} \pm 1)^2$ gives

$$\log\left(q + \frac{1}{2} \pm \sqrt{2q}\right) = h \pm d, \qquad 0 < d < h.$$

The elementary inequalities

$$\sqrt{1+z} + \sqrt{1-z} \le 2 - \frac{z^2}{4}, \qquad \log\frac{1+z}{1-z} \ge 2z \qquad (0 \le z < 1)$$

follow, respectively, by squaring and by integrating $2/(1 - z^2) \ge 2$. In particular, $d \ge \sqrt{2/q}$. The angular comparison therefore gives

$$\begin{aligned} \frac{L\phi(x)}{\phi(x)} &\ge 4 - 2\sqrt{1 + d/h} - 2\sqrt{1 - d/h} \\ &\ge \frac{d^2}{2h^2} \ge \frac{1}{qh^2} > \frac{1}{q(\log q)^2} = W(x). \end{aligned}$$

This proves (2.4).

It remains to check $q = 2, 4, 5$. Write $W_q = 1/[q(\log q)^2]$. We use the rational bounds

$$\frac{18}{25} < F(4), \qquad \frac{173}{200} < F(5) < \frac{87}{100},$$

$$F(8) < \frac{201}{200}, \qquad F(9) < \frac{26}{25}, \qquad F(10) < \frac{213}{200}, \tag{2.5}$$

$$W_2 < \frac{21}{20}, \qquad W_4 < \frac{2}{15}, \qquad W_5 < \frac{2}{25}. \tag{2.6}$$

All these bounds follow by substitution in the elementary enclosure

$$0 < \log t - 2\sum_{k=0}^{4} \frac{z^{2k+1}}{2k+1} < \frac{2z^{11}}{11(1-z^2)}, \qquad z = \frac{t-1}{t+1}, \quad t > 1. \tag{2.7}$$

To obtain it, expand $\log t = 2\sum_{k\geq 0} z^{2k+1}/(2k+1)$ and bound the tail by a geometric series. Substituting (2.5)–(2.6) into the three neighbor sums gives

$$\begin{aligned} (L-W)\phi(1,1) &= (4-W_2)\frac{3}{5} - 2F(5) > \frac{3}{100}, \\ (L-W)\phi(2,0) &= (4-W_4)\frac{18}{25} - F(9) - 2F(5) > \frac{1}{250}, \\ (L-W)\phi(2,1) &= (4-W_5)F(5) - \frac{3}{5} - \frac{18}{25} - F(8) - F(10) > \frac{1}{1250}. \end{aligned} \tag{2.8}$$

Sign changes and coordinate permutations cover all remaining points. □

Proof of the lower bound in Theorem 1.1. Apply (2.1) to the function (2.3). The first term is nonnegative, and Proposition 2.3 gives $L\phi/\phi > W$ at every point of D. Hence $\mathcal{E}(u) \geq \sum_D W(x)u(x)^2$, with strict inequality for $u \neq 0$. □

The same estimates give an explicit remainder.

Corollary 2.4. *For $x \in D$, put $q = |x|^2$ and*

$$R(x) = \begin{cases} \dfrac{1}{1250}, & 1 < q < 8, \\ \dfrac{1}{q\log^2(q-\frac{1}{2})} - \dfrac{1}{q(\log q)^2}, & q \geq 8. \end{cases}$$

Then $R(x) > 0$ on D, and every $u \in C_c(D)$ satisfies

$$\mathcal{E}(u) \geq \frac{1}{4}\sum_{x\in D} \frac{|u(x)|^2}{|x|^2(\log|x|)^2} + \sum_{x\in D} R(x)|u(x)|^2. \tag{2.9}$$

Proof. For $1 < q < 8$, we have $\phi(x) < 1$, so (2.8) gives $(L-W)\phi(x)/\phi(x) > 1/1250$. For $q \geq 8$, use (2.4). The conclusion follows from (2.1). □

The coefficient $1/4$ is therefore sharp within the family of constant multiples of the prescribed weight, although the weight admits a positive additive improvement. No optimality is asserted for the remainder R.

3. Sharpness

Choose $\chi \in C^\infty([0,\infty))$ with $0 \leq \chi \leq 1$, $\chi = 1$ on $[0,1]$, and $\chi = 0$ on $[2,\infty)$. For $T \geq 2$, put

$$u_T(x) = \begin{cases} \sqrt{\log|x|}\,\chi(T^{-1}\log|x|), & x \in D, \\ 0, & x \in K. \end{cases}$$

These functions are finitely supported. Write

$$\mathcal{V}_b(u) = \sum_{x\in D} \frac{|u(x)|^2}{|x|^2\log^2(b|x|)} \quad (b \geq 1), \qquad \mathcal{V} = \mathcal{V}_1.$$

Let $U_T(z) = \sqrt{\log|z|}\,\chi(T^{-1}\log|z|)$ for $|z| > 1$. On $r = |z| \geq 2$, differentiation gives, uniformly in T,

$$|\nabla U_T(z)| \leq Cr^{-1}, \qquad |\nabla^2 U_T(z)| \leq Cr^{-2}.$$

The gradients of $|\nabla U_T|^2$ and $U_T^2/(r^2\log^2 r)$ are consequently bounded by Cr^{-3}. Taylor's theorem and comparison on unit squares therefore give uniformly bounded errors when replacing lattice

sums by integrals outside a fixed ball, since $\sum_{|x|\geq 3} |x|^{-3} < \infty$. The finitely many remaining lattice terms are also uniformly bounded.

With $t = \log r$, it follows that

$$\begin{aligned}\mathcal{V}(u_T) &= 2\pi \int_{\log 2}^{\infty} \frac{\chi(t/T)^2}{t}\, dt + O(1) = 2\pi \log T + O(1),\\ \mathcal{E}(u_T) &= 2\pi \int_{\log 2}^{\infty} \left(\frac{\chi(t/T)}{2\sqrt{t}} + \frac{\sqrt{t}}{T}\chi'(t/T) \right)^2 dt + O(1)\\ &= \frac{\pi}{2} \log T + O(1).\end{aligned}$$

Indeed, $\chi' = 0$ unless $T \leq t \leq 2T$, so the terms containing χ' contribute $O(1)$. Hence $\mathcal{E}(u_T)/\mathcal{V}(u_T) \to 1/4$, proving sharpness in Theorem 1.1.

For fixed $b \geq 1$, the bound $u_T(x)^2 \leq \log |x|$ gives

$$0 \leq \mathcal{V}(u_T) - \mathcal{V}_b(u_T) \leq 2(\log b) \sum_{x \in D} \frac{1}{|x|^2 (\log |x|)^2} = O_b(1).$$

The series converges, so $\mathcal{E}(u_T)/\mathcal{V}_b(u_T) \to 1/4$, proving Corollary 1.2. Removing any fixed finite set from the weighted sums changes them by $O(1)$; the coefficient remains sharp after such a removal.

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